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Database Foundations

3-1

More with Relationships

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Objectives

- This lesson covers the following objectives:
 - Identifying barred relationships
 - Resolve M:M relationships
 - Identify and illustrate nontransferable relationships
 - Identify and draw supertype and subtype entities
 - Identify hierarchical, recursive, and arc relationships



Identifying (Barred) Relationships

- What is the UID of the ACCOUNT entity?



- The UID requires both the ACCOUNT Number and the relationship between BANK and ACCOUNT



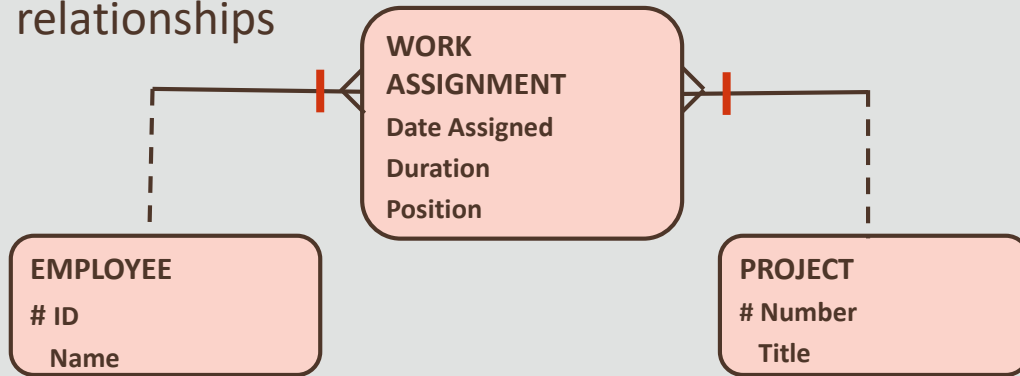
- The UID for the ACCOUNT entity is made up of ACCOUNT Number as well as BANK Number – this is represented by the bar on the relationship line

An identifying relationship is created when the UID for an entity includes the relationship with another entity.

Note that a relationship included in a UID must be mandatory and one-and-only-one in the direction that participates in the UID.

Identifying Relationships with Multiple Entities

- An entity can be uniquely identified through multiple relationships



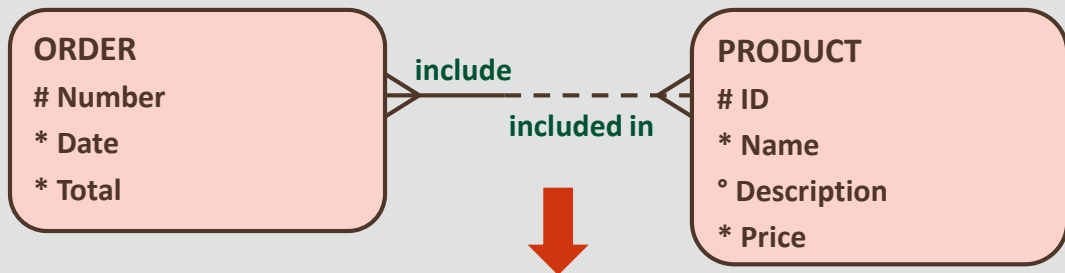
- The UID for WORK ASSIGNMENT is EMPLOYEE ID and PROJECT Number (represented using bars)

In the slide example, an EMPLOYEE and a PROJECT are needed to make WORK ASSIGNMENT unique, so both relationships are included in the UID for WORK ASSIGNMENT.

Note that WORK ASSIGNMENT is an intersection entity that is the resolution between an M:M relationship. This topic is discussed in a later lesson.

M:M Relationships

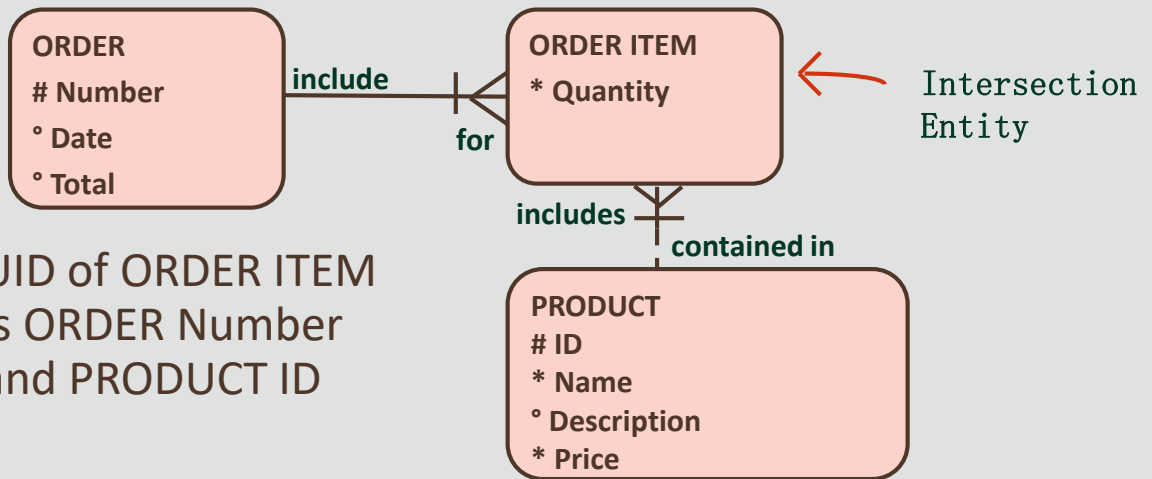
- Attributes describe only entities
- If attributes describe a relationship, the relationship must be resolved



Where would you add the Quantity attribute?

Resolving M:M Relationships: Example 1

- Resolving a M:M relationship with a new intersection entity and two 1:M barred relationships



- UID of ORDER ITEM is ORDER Number and PRODUCT ID

Intersection entity characteristics:

- The relationships from the intersection entity are always mandatory
- Intersection entities usually contain consumables like quantity used and dates. They tend to be high-volume and volatile entities
- An intersection entity is identified by its two originating relationships (identifying relationships)

Note: Choosing a name for the intersection entity is often difficult. Leave it blank at first, move the relationships, add the attributes, and then it will often be clearer what the name should be.

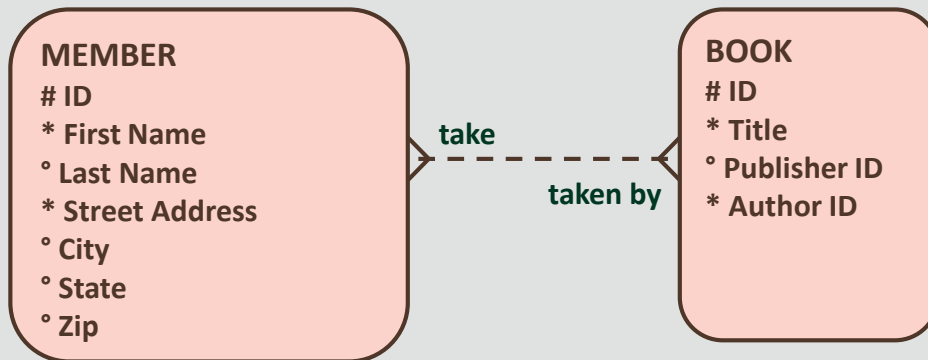
If you do not have any additional attributes in the intersection entity, you can leave it as an M:M relationship and Oracle SQL Developer Data Modeler will create the intersection table in the relational model.

Case Scenario: Resolving M:M Relationships



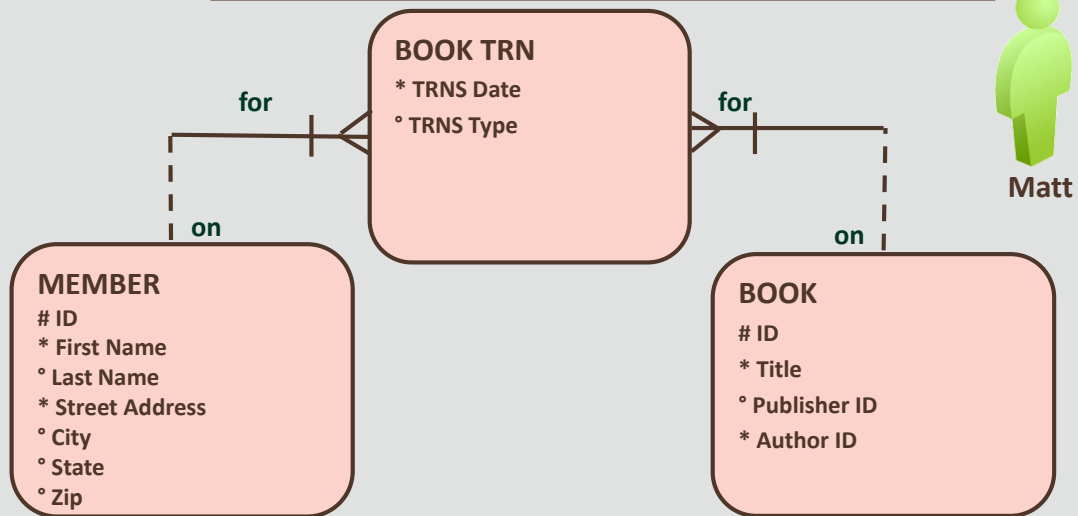
Faculty

Matt, how would you resolve the following M:M relationship between the MEMBER and the BOOK entities?



Case Scenario: Creating an Intersection Entity

I would create an intersection entity with identifying relationships to the originating entities.



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A third entity, called an intersection entity, is required to resolve the M:M relationship.

To resolve the M:M relationship from the previous slide, perform the following steps:

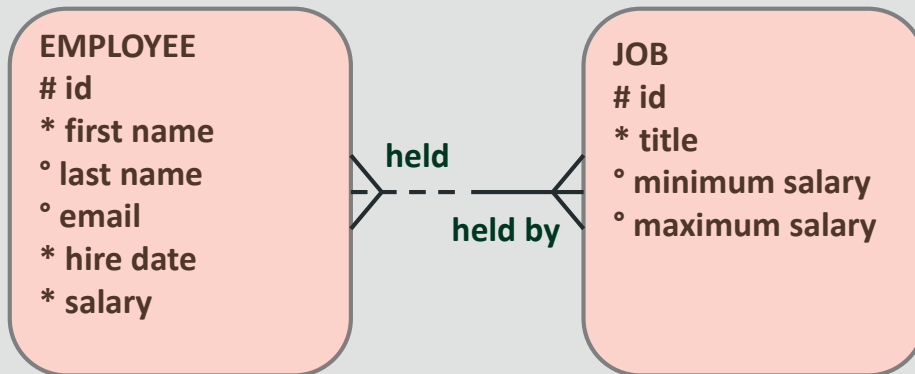
- Delete the M:M relationship.
- Create a new entity called **BOOK TRN**.
- Create two identifying relationships: one from **MEMBER** and the other from **BOOK**.
- Identify and create additional attributes in the **BOOK TRANSACTION** intersection entity.
- Evaluate whether the two identifying relationships constitute a UID for the intersection entity or whether other attributes are needed.

TRNS ID is an artificial UID that could be used instead of barred relationships. It is an alternative method, not better or worse.

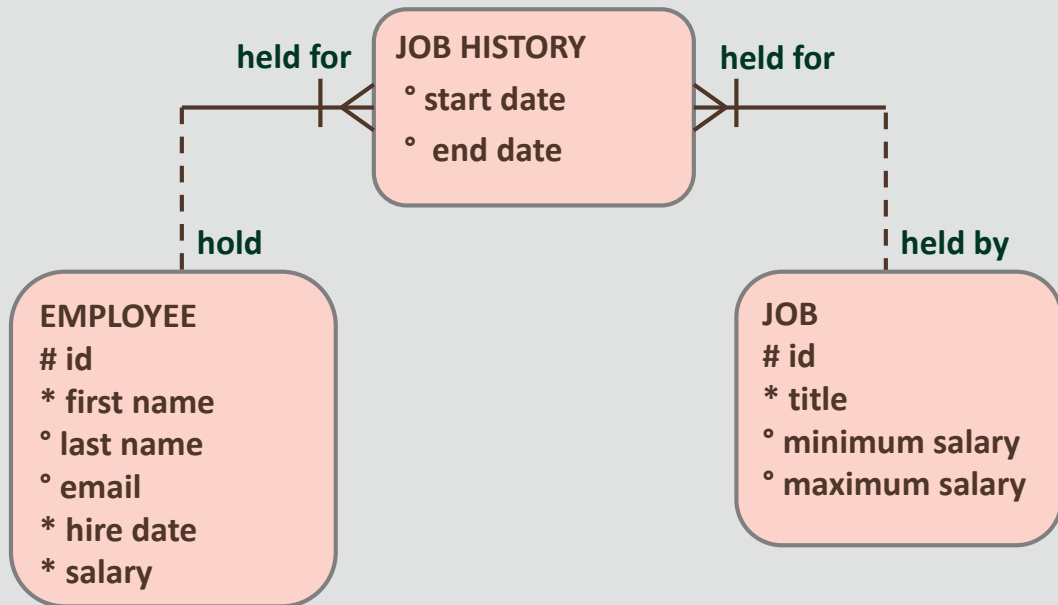
Note: Not all intersection entities will have additional attributes.

Resolving M:M Relationships: Example 2

- The EMPLOYEE and JOB entities do not store the history of an employee's jobs



Resolving M:M Relationships Using Intersection Entity with Barred Relationships



There is a M:M relationship between EMPLOYEE and JOB. There is no way to store information about an employee's jobs. The JOB HISTORY intersection entity stores this information with two identifying relationships for the EMPLOYEE and JOB entities.

A barred relationship is mapped to a foreign key column on the many side, just like any other M:1 relationship. The UID of the intersection entity often comes from the originating relationships and is represented by bars.

Project Exercise 1

- DFo_3_1_1_Project
 - Oracle Baseball League Store Database
 - Resolving Many to Many Relationships



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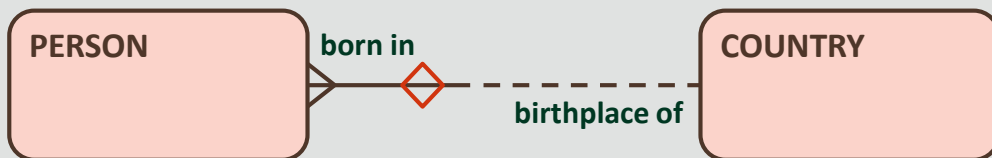
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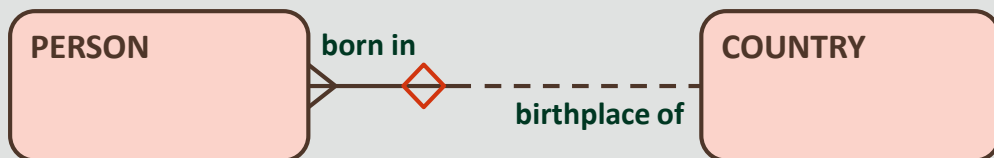
Non-Transferable Relationships

- Transferability is the ability of the relationship between two instances of an entity to change over time
- A non-transferable relationship cannot be moved between instances of the entities it connects
- A non-transferable relationship is represented by a diamond on the relationship



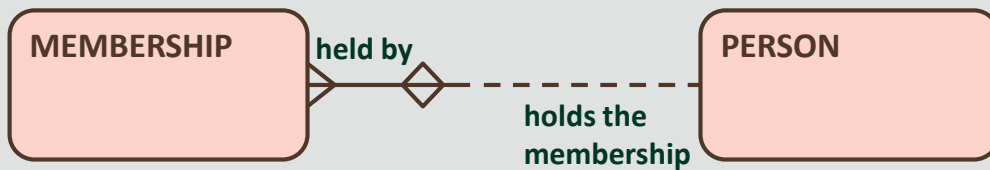
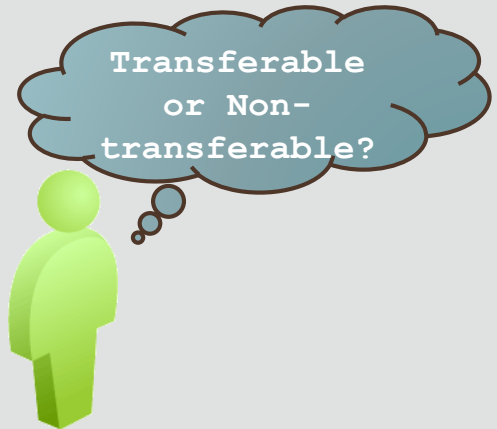
Non-Transferable Relationships

- Non-transferable relationships can only be mandatory
- For example, the birth country of a person is non-transferable



Case Scenario

- A membership must be held by exactly one person
- The membership relationship cannot be moved to another person



Project Exercise 2

- DFo_3_1_2_Project
 - Oracle Baseball League Store Database
 - Identify and Illustrate Non-Transferable Relationships



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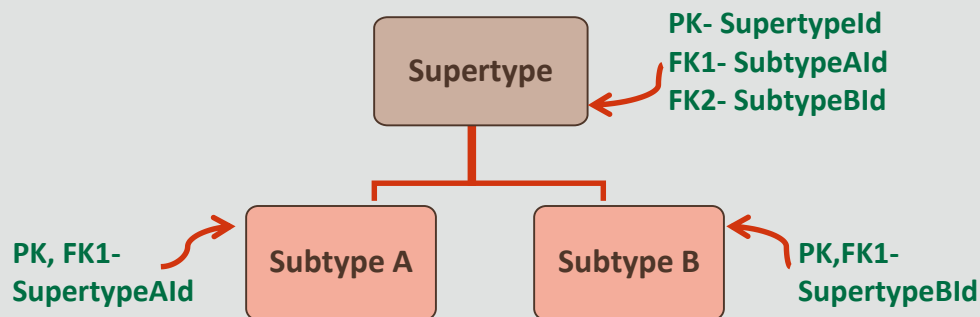
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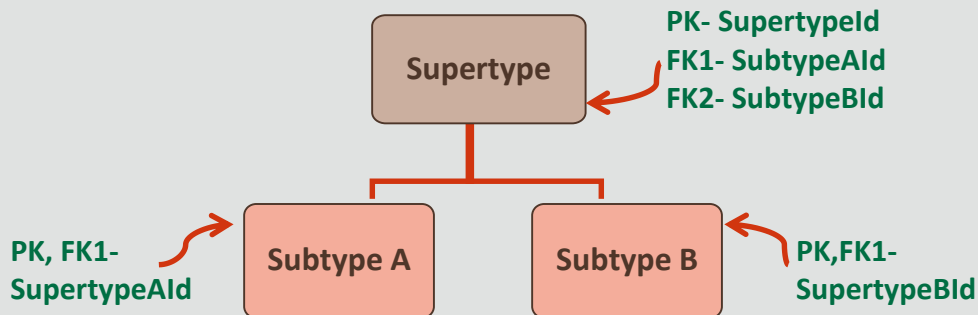
Supertype and Subtype Entities

- Supertype has a parent-child relationship with one or more subtypes



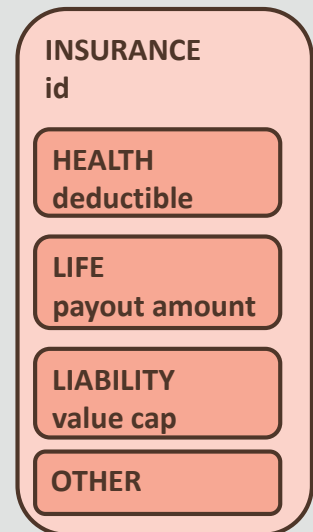
Supertype and Subtype Entities

- Subtype is a subgrouping of the entity in an entity type which has attributes that are distinct from those in other subgroupings



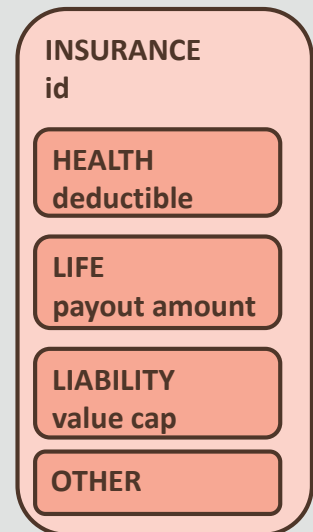
Drawing a Subtype

- Each subtype is a specialization of a supertype and therefore must be enclosed within an entity
- The common attributes and relationships for all subtypes must be listed only in the supertype, but they are inherited in every subtype



Drawing a Subtype

- A subtype can and would generally have attributes and relationships of its own
- There can never be just one subtype; another subtype should be created to contain the rest



Characteristics of a Subtype

- A subtype:
 - Inherits all attributes of the supertype
 - Inherits all relationships of the supertype
 - Usually has its own attributes or relationships
 - Is drawn within the supertype
 - Never exists alone
 - May have subtypes of its own
 - Has identical primary keys of the supertype and subtype

Supertype and Subtype Entity: Example



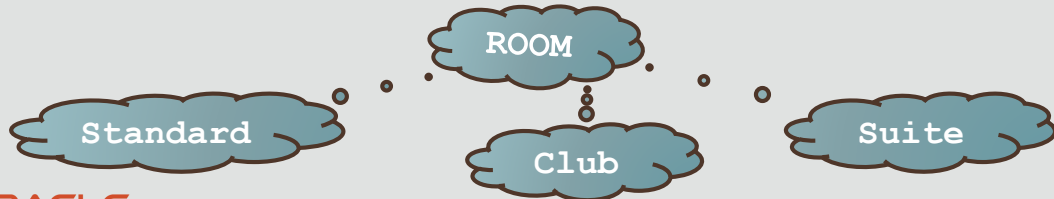
Sharon wants to include types of room categories to be reserved for the guests. For example:

Standard

Club

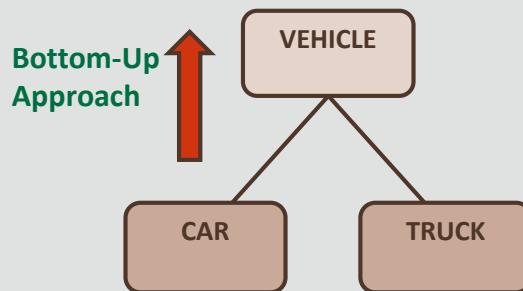
Suite

She already has an entity named ROOM that holds room details. This would be a supertype entity. The specific room categories would inherit the properties of the supertype entity, in addition to their own specific attributes. The specific room category would be the subtype entity.



Generalization and Specialization

- Generalization is a bottom-up approach where two or more lower level entities are combined to form a higher level entity based on the common feature

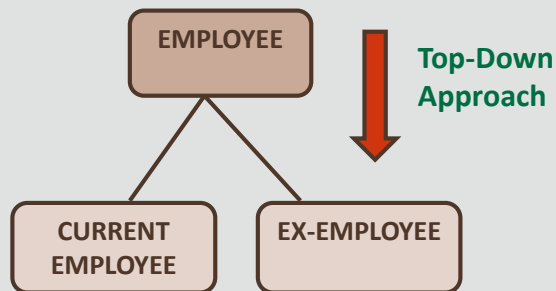


Generalization is the process of extracting shared characteristics from two or more entities, and combining them into a generalized supertype. Shared characteristics can be attributes, associations, or methods.

In the slide example, CAR and TRUCK share the same common attributes, and so their common attributes are grouped to form the supertype entity named Vehicle.

Generalization and Specialization

- Specialization is a top-down approach where the higher level entity is broken down into lower level entities



Specialization means creating new subtypes from an existing entity. If it turns out that certain attributes, associations, or methods apply only to some of the entity's objects, a subtype can be created.

In the slide example, the EMPLOYEE supertype is broken down into two subtypes, CURRENT EMPLOYEE and EX-EMPLOYEE.

Entity Subtype Rules

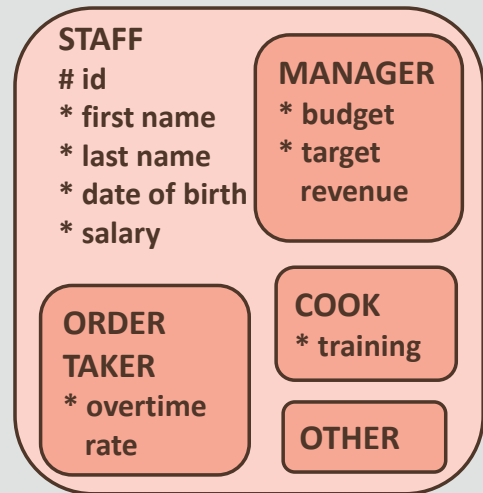
- Exhaustive:
 - Every instance of a supertype is also an instance of one of the subtypes
 - OTHER should be included as a subtype to categorize instances that are not defined by one of the existing subtypes
 - Example: An employee must be full time, part time, or other

Entity Subtype Rules

- Mutually Exclusive:
 - Every instance of the supertype is of one and only one subtype
 - Example: An employee cannot be both full time and part time

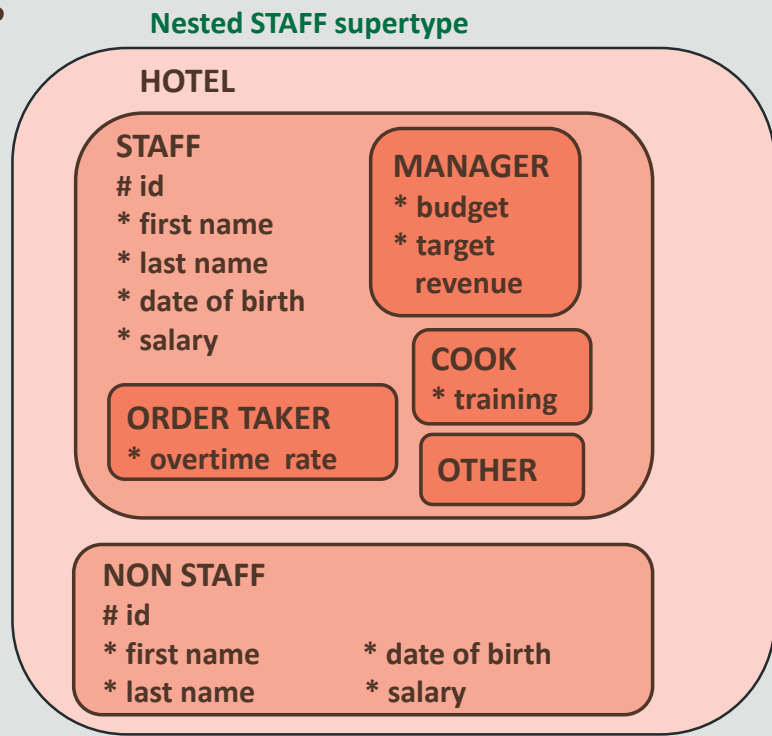
Identifying Subtypes Correctly

- Is this subtype a kind of supertype?
- Have I covered all possible cases? (exhaustive)
- Does each instance fit into one and only one subtype? (mutually exclusive)



Nested Subtypes

- You can nest subtypes



Project Exercise 3

- DFo_3_1_3_Project
 - Oracle Baseball League Store Database
 - Identify and Draw Supertype and Subtype Entities



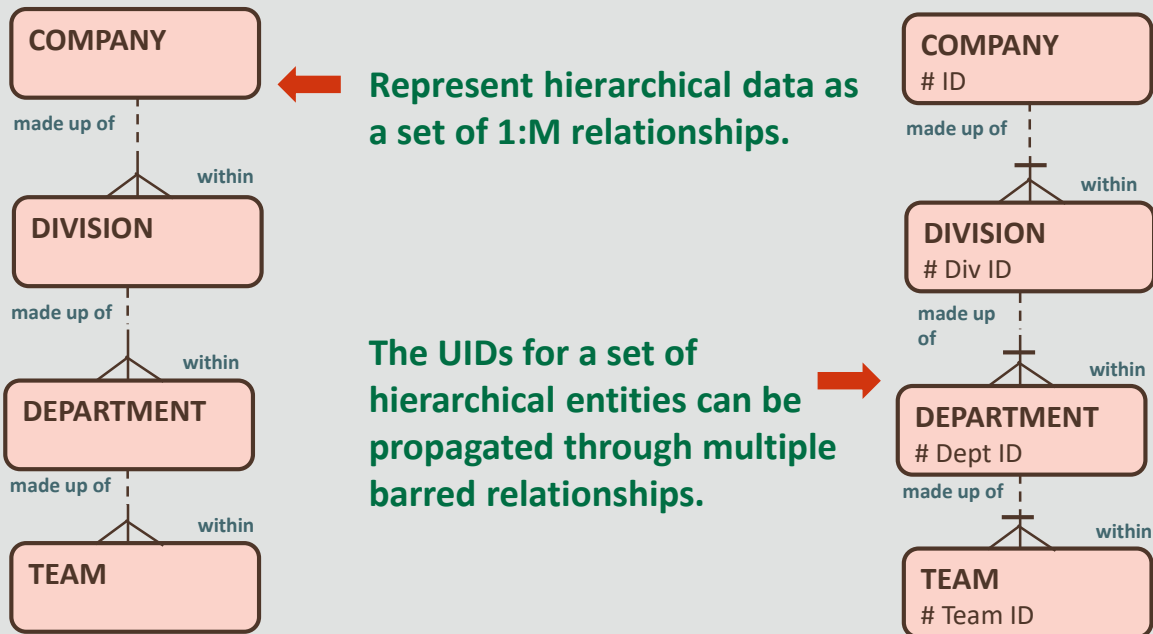
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Modeling Hierarchical Data



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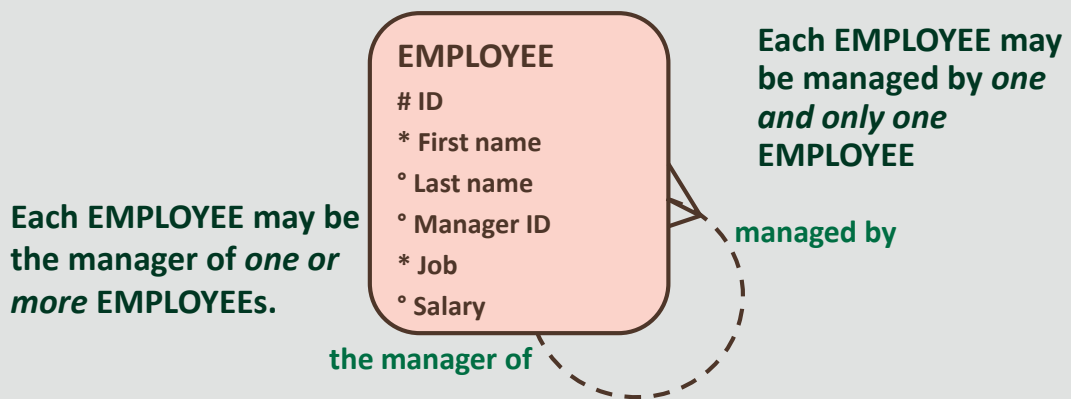
One technique for modeling relationships is to create a set of M:1 relationships hierarchically. In the slide example, an organization has a hierarchical structure:

- A COMPANY may consist of one or more DIVISIONs.
- A DIVISION may consist of one or more DEPARTMENTs.
- A DEPARTMENT may consist of one or more TEAMs.

The UIDs for a set of hierarchical entities may be propagated through multiple relationships. Consider creating artificial attributes to help identify entities in a hierarchical relationship.

Recursive Relationships

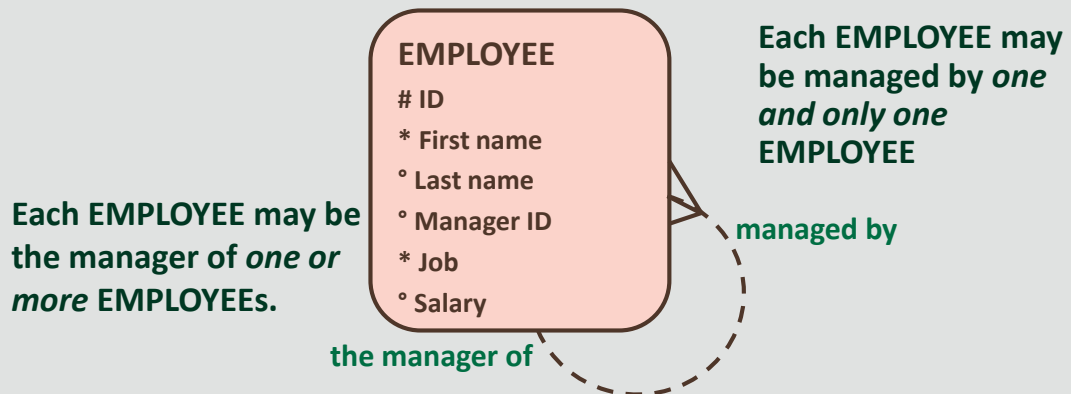
- A recursive relationship is one where an entity instance is related to another instance in the same entity



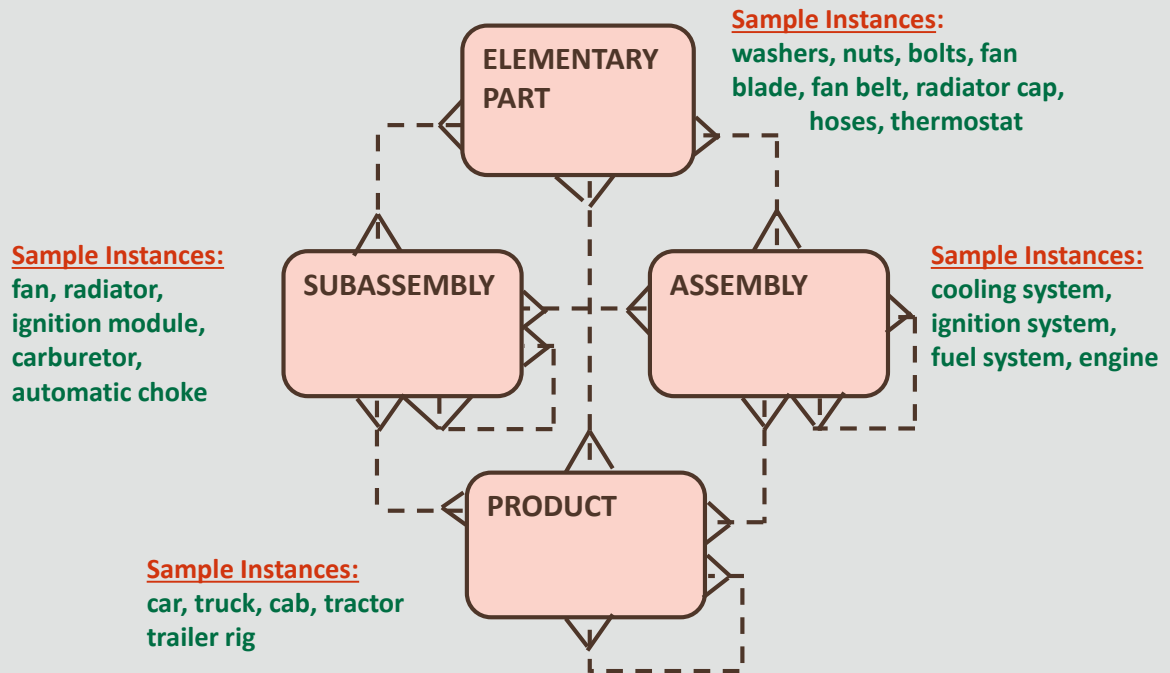
In this example, every manager is also an employee. So manager is not a new entity, but just a subset of the instances of the EMPLOYEE entity. This many-to-one relationship is optional in both directions. It is the same as the relationship between two distinct entities.

Recursive Relationships

- A recursive relationship is always modeled with a loop



Examining Recursive Relationships



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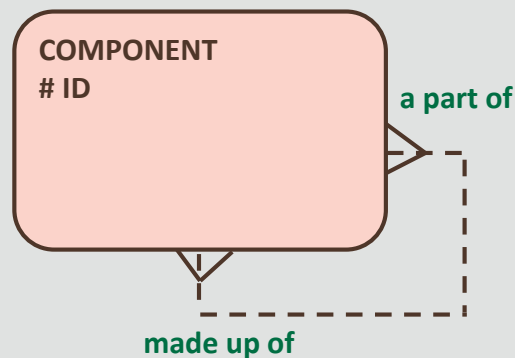
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An automobile manufacturing organization needs to track elementary parts, subassemblies, assemblies, and products. Bill of Materials data can be modeled with multiple entities for each category of "part" and a set of relationships between each of those entities.

The ERD in the slide depicts the data by considering each of these part categories as an entity.

Generic Modeling

- Another way to model a Bill of Materials recursive relationship is to create a generic COMPONENT entity
- This would create a recursive M:M relationship



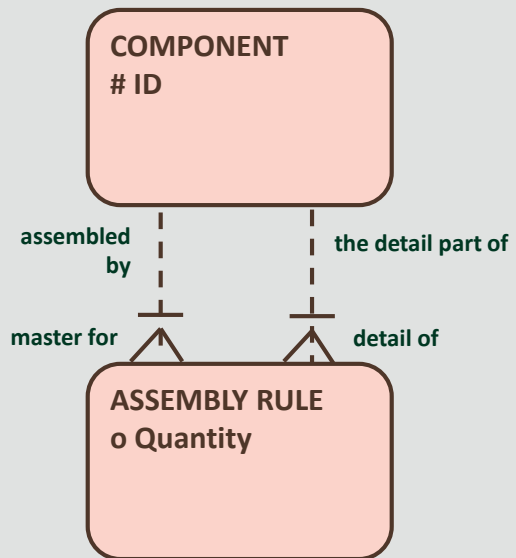
Consider all elementary parts, subassemblies, assemblies, and products as instances of an entity called COMPONENT. Then the complex ERD on the previous page can be remodeled as a simple recursive relationship.

Suppose that some of the instance data supplied in the previous slide is used to validate this model:

- Washers, nuts, and bolts may be part of various SUBASSEMBLIES, ASSEMBLIES, and PRODUCTS.
- A fan blade and a fan belt are part of a fan.
- A radiator cap, host, and thermostat are part of a radiator.
- A cooling system, ignition system, and engine block are part of a complete engine.

Resolving a M:M Recursive Relationship

- Resolve the M:M recursive relationship with an intersection entity



The attribute quantity is associated with the recursive relationship, and therefore must be resolved. Resolve the M:M recursive relationship by adding the ASSEMBLY RULE intersection entity and two 1:M relationships back to the COMPONENT entity. ASSEMBLY RULE will have a Quantity attribute.

Using the instance data again, the ASSEMBLY RULE instance for washers to fan will have a 1:M relationship to the COMPONENT instance for washer and a second 1:M relationship to the COMPONENT instance for fan. The ASSEMBLY RULE entity will record the quantity of washers that are part of a single fan.

Arc Relationship

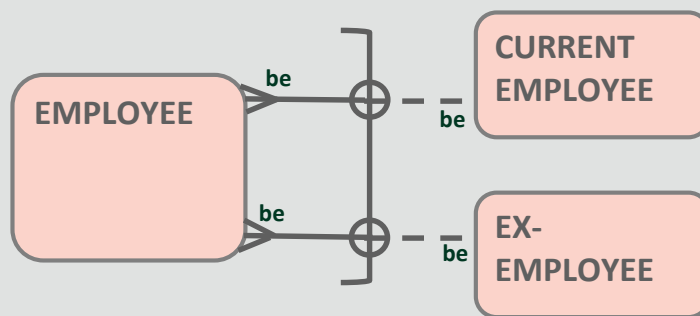
- An arc is an exclusive relationship group, which is defined such that only one of the relationships can exist for any instance of an entity
- The arc drawn between two relationships connects them and demonstrates mutual exclusivity
- The relationship implies an "or" condition
- The arc indicates that any instance of that entity can have only one valid relationship of the relationship in the arc at any one time

Characteristics of an Arc Relationship

- The relationships in an arc frequently have the same relationship name
- The relationships in an arc must be either all mandatory or all optional and should have the same cardinality
- An arc belongs to a single entity and must include only relationships that originate from the entity
- An entity may have multiple arcs, but a specific relationship can participate only in a single arc
- An arc relationship is represented as the arc-shaped line across two or more relationship lines
 - Relationships included in the arc have a circle on the relationship arc line

Arc Relationships

- A supertype entity and its subtypes can be modeled as an arc relationship
- Example: An EMPLOYEE entity is either a CURRENT EMPLOYEE or an EX-EMPLOYEE, but not both (this could also be modeled as a supertype/subtype)



The supertype and subtype (arc) implementation produces one table for every entity. The supertype table has a foreign key for each subtype table. These foreign keys represent exclusive relationships and are optional because only one foreign key can have a value for each row in the table.

Case Scenario: Arc Relationship



Faculty

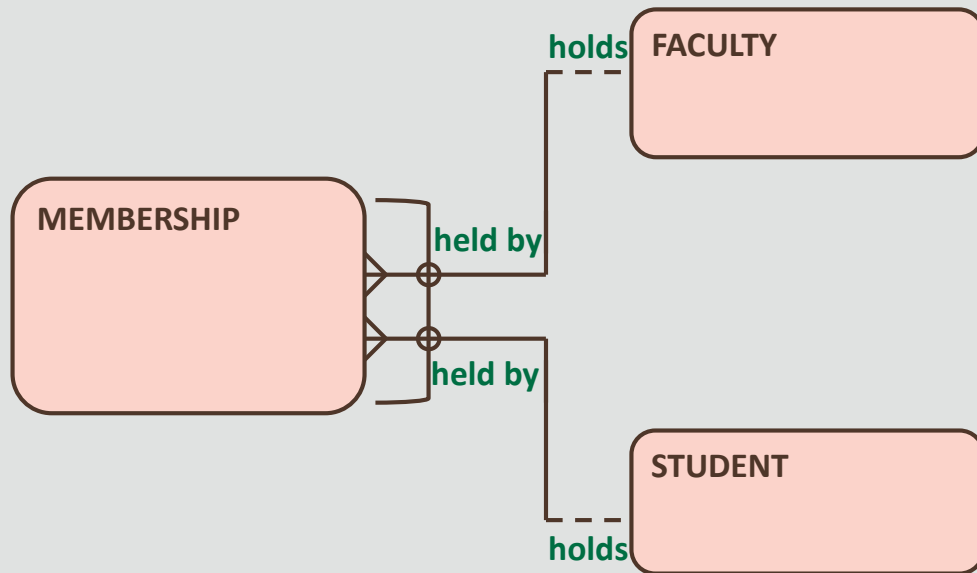
Matt, can you create an entity that can have an arc relationship?

**I can create a common entity called
MEMBERSHIP that would hold
membership details common to all
membership categories**



Matt

Case Scenario: Creating a Common Entity



Each MEMBERSHIP may be held by one and only one FACULTY or it may be held by one and only one STUDENT.

Each FACULTY may be the holder of one or more MEMBERSHIPS. Each STUDENT may be the holder of one or more MEMBERSHIPS. A MEMBERSHIP may not be held by both a FACULTY and a STUDENT. This mutual exclusivity is modeled by the arc.

Arcs may be drawn for optional relationships and for mandatory relationships, but the two relationships participating in the arc must have the same cardinality.

Project Exercise 4

- DFo_3_1_4_Project
 - Oracle Baseball League Store Database
 - Identify Hierarchical, Recursive and Arc Relationships



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Summary

- In this lesson, you should have learned how to:
 - Identifying barred relationships
 - Resolve M:M relationships
 - Identify and illustrate nontransferable relationships
 - Identify and draw supertype and subtype entities
 - Identify hierarchical, recursive, and arc relationships



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